17. A farm machinery manufacturing company has designed a new seed drill. In order to study the efficiency of the seed drill as compared to the existing one, an experiment was carried out with plots of land grouped in to 12 pairs. One of them was sown using special drill and the other by ordinary drill. The grain yields are given as follows: The company claims that their new seed drill increases the yield. Use paired t test, examine whether the data provide an evidence for or against their claim.

Plot pair no	Special drill	Ordinary drill
1	7.9	6.5
2	8.4	8.0
3	6.4	6.4
4	8.6	7.5
5	5.7	5.6
6	5.5	6.2
7	5.9	6.4
8	7.3	6.9
9	6.1	6.8
10	8.3	7.7
11	7.3	6.0
12	6.9	6.2

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18. The hours of spending in front of TV and test paper marks of 10 school students are as follows: Find Pearson's coefficient of correlation and interpret the result. Also represent the data by means of a scatter diagram.

TV hours	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5
TP marks	89	78	75	56	55	52	44	37	31	25

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19. The following data relates to prices of 10 commodities and quantities demanded. Find Pearson's coefficient of correlation between price and demand and interpret the result. Also represent the data by means of a scatter diagram.

Price	10	14	15	30	25	27	32	18	50	5
Demand	68	19	37	46	55	52	44	37	10	10

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20. The following data provides the mental age(x) and scores in a test (y) of a group of 10 students in a class. Calculate Karl Pearson correlation coefficient between ages (x) and scores (y) of students and interpret the result. Also represent the data by means of a scatter diagram.

age	10	12	13	14	16	17	18	5	7	9
score	24	20	23	24	18	19	20	1	8	20

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21. The following data provides the length (x) in cm and weight (y) in gms of a sample of 10 articles in a factory. Calculate Karl Pearson correlation coefficient between length (x) and weight (y) of articles and interpret the result. Also represent the data by means of a scatter diagram.

length	4	12	10	20	16	9	10	9	12	3
weight	19	10	13	24	18	19	12	16	14	8

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22. The following data relates to marital status and performance in an examination. Use chi-square test, examine whether the performance depends on marital status and interpret the result.

	performance	
	Good	Bad
married	70	90
unmarried	30	50

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23. The following table shows the association among 1000 school boys of their general ability and their mathematical ability. Use chi-square test, examine whether general ability and mathematical ability are independent.

	General ability			
		Good	Fair	Poor
Mathematical ability	Good	57	20	3
	Fair	260	251	154
	Poor	59	94	102

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24. The following data gives the yield of wheat in 30 test plots which are given 3 different fertilizers: Use ANOVA, test whether the fertilizers are equal in their effects.

Fertilizer1	80	70	65	60	60	50	75	80	75	85
Fertilizer2	80	75	70	65	60	60	70	75	80	85
Fertilizer3	60	60	60	50	50	40	65	75	70	70

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25. The varieties A, B, C, D, E, F, G of paddy were sown in 5 plots each and the following yields in quintals per acre were obtained. Use ANOVA, test whether the data indicate a significant difference in the yields of the varieties.

A	B	C	D	E	F	G
13	10	12	15	13	13	14
12	12	11	10	10	11	12
18	16	13	17	14	15	19
15	13	14	14	15	16	17
11	11	10	10	9	12	15

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